

Workbook: Blood

Week 3 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on blood: its composition, the formed elements (red cells, white cells, platelets), plasma and its proteins, blood typing, and basic clinical anatomy of common blood disorders.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Blood as a tissue · 3 questions

Q01 · DOK 1 · Recall

Why is blood classified as a connective tissue?

Q02 · DOK 1 · Recall

Approximately what fraction of body weight is blood, and what is the rough total volume in an adult?

Q03 · DOK 2 · Apply

What is hematocrit, and what is a normal value?

Plasma · 2 questions

Q04 • DOK 1 • Recall

What is plasma made of?

Q05 • DOK 1 • Recall

Name the three major classes of plasma proteins and what each does.

Red blood cells • 3 questions**Q06 • DOK 1 • Recall**

Describe the structure of a red blood cell and how its structure suits its function.

Q07 • DOK 1 • Recall

What is hemoglobin, what is it made of, and what does it do?

Q08 • DOK 1 • Recall

How long do red blood cells live, and where are they made and broken down?

White blood cells • 5 questions

Q09 · DOK 1 · Recall

Name the five types of white blood cells in order of typical abundance and group them as granulocytes or agranulocytes.

Q10 · DOK 1 · Recall

What do neutrophils do, and what is their nucleus like?

Q11 · DOK 2 · Compare

Compare eosinophils and basophils.

Q12 · DOK 1 · Recall

What do lymphocytes do, and what are the major subtypes?

Q13 · DOK 1 · Recall

What do monocytes do, and what is their nucleus like?

Platelets and hemostasis · 2 questions

Q14 · DOK 1 · Recall

What are platelets, what cell are they from, and what do they do?

Q15 · DOK 2 · Apply

Outline the three general steps of hemostasis after a vessel is cut.

Blood types · 3 questions

Q16 · DOK 1 · Recall

What does the ABO blood type system describe, and what are the four ABO types?

Q17 · DOK 1 · Recall

What does the Rh blood type system describe?

Q18 · DOK 3 · Reason

Type O-negative blood is called the 'universal donor' for emergency transfusion. Why?